

UNIVERSITÀ DEGLI STUDI DI MILANO-BICOCCA

Scuola di Scienze

Dipartimento di Fisica

Master's degree in Artificial Intelligence for Science and Technology



Book Recommendation System Using the Children's Books Dataset

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1. Abstract

The goal of this project is to use information from the Goodreads dataset to create a personalized recommendation system for children's books. The main goal is to assess and improve the Neural Matrix Factorization (NeuMF) model by implementing two significant enhancements: attention mechanisms and transfer learning. Whether these improvements may raise recommendation quality as determined by ranking measures is the main study question.

There are four steps in our process. To set a performance standard, we first implement the NeuMF architecture. To provide richer initial representations, we then incorporate transfer learning by utilizing pretrained embeddings. To enhance the model's capacity to capture user preferences, the third model adds an attention layer that weighs latent features selectively. To investigate their potential synergy, we finally integrate both methods into a single design.

Using two commonly used ranking metrics, Hit Rate at 10 (HR@10) and Normalized Discounted Cumulative Gain at 10 (NDCG@10), we assess the models for a range of factor sizes (8, 16, 32, 64). According to preliminary findings, both transfer learning and attention mechanisms enhance performance, with the combined model typically obtaining maximum accuracy. This study shows how neural improvements can be used to improve collaborative filtering for book recommendation systems.

2. Introduction

Due to the excessive amount of information on the internet, personal discovery becomes important. On social media, streaming, and shopping platforms, recommendation systems are now essential tools for guiding users toward content that best suits their interests. To make predictions about preferences and increase engagement, these systems study item attributes and user behavior. Collaborative filtering has grown more popular by being able to recognize patterns of user-item interaction. Deep learning has come along, and these algorithms can now detect complex, non-linear patterns with much better-quality suggestions.

This study aims to create a complex book recommendation system using the Children's Books dataset from Goodreads. Non-interactions in this data cannot be interpreted as disinterest, only positive interactions (like ratings) are observed. The data contains implicit feedback, which makes this a major challenge. The study frames the prediction as a binary classification problem and generates training data using negative sampling to get around this. Despite the sparsity and implicit nature of the dataset, the primary objective is to provide accurate, customized suggestions.

First, the project will deploy a Neuralized Matrix Factorization (NeuMF) baseline model by combining General Matrix Factorization (GMF) and Multi-Layer Perceptron (MLP) to leverage both linear and non-linear information. Secondly, to reduce data sparsity and enhance model generalizability, the research leverages pre-trained embeddings and transfer learning. Third, it examines the use of attention mechanisms to enable the model to closely focus its attention on the most important latent features, leading to more accurate predictions. Lastly, it attempts to combine both transfer learning and attention in one model to determine their combined influence on recommendation quality.

This research is significant because it provides a methodical assessment of cutting-edge neural recommendation methods on a real-world dataset. It offers valuable insight into the performance of the algorithms in environments with implicit and sparse feedback. Furthermore, the results can inform the creation of more effective and efficient recommendation systems, which will lead to increased user satisfaction and content discovery on digital platforms.

3. Materials and Methods

The basis for this project is the data set for children's books, which is taken from the large Goodreads data set available online at <https://cseweb.ucsd.edu/~jmcauley/datasets/goodreads.html> . It involves user -based book assessment, and acts as an underlying response data setting of positive interactions, such as rankings, clearly without marking the negative clearly.

The workflow followed the five main steps: data processing, preparation of a dataset for model training, production of model architecture, training models and eventually evaluating their performance.

Data Preprocessing

The original dataset came as a compressed Json file. Each row was its own Json object. A python script treated each line to remove user -D, book -D and ranking. Entries were placed with only 1 or higher rankings, and the result was stored in a space-practical format (user_id book_id-rating) called test.data.

To make the dataset more manageable, further filtration was used. Users who evaluated 5 books and books with less than 20 rankings were removed. To speed up growth, the dataset was then downsampled to just 20% of the filtered data.

Dataset preparation

A customized PyTorch dataset class, NCF_Data, was written to handle user and item ID encoding, follow each user's positive ranked items, and share the data using a leave-one-out strategy. In this setup, we held out each user's single highest-rating interaction for testing and used all other ratings for training., while the rest was used for training.

Since the dataset only included positive responses, negative samples were artificially generated. For each positive training, four negative samples were created using items the user hadn't interacted with. For testing, 99 negative items were sample per user, which was paired with their single test elements. It created a pool with 100 options to rank the model, which simulates a real-world recommendation task. DataLoaders were then designed to feed these batches into the model efficiently,, using a batch size of 256.

Model architecture

Four models were made based on the Neural Collaborative Filtering (NCF) framework:

GMF (General Matrix Factorization): Used an elemental product to predict interaction by learning embeddings for users and elements. Embedding sizes tested including 8, 16, 32 and 64.

MLP (Multi-Layer Perceptron): Combined user and item embeddings through concatenation and passed them through multiple dense layers with ReLU activations. Embedding sizes and layer configuration were varied.

NeuMF: Combined GMF and MLP into one model. The outputs of both models were merged and passed through a final prediction layer. Pre-trained GMF and MLP models could optionally be used here.

NeuMF with Transformer: Built on top of NeuMF by adding a Transformer encoder block, allowing the model to better learn complex relationships. The embedding dimensions for GMF and MLP were 16 and 32 respectively, and the transformer input was set at 64.

Training

All models used binary cross-entropy to evaluate performance and were optimized with Adam Optimizer (Learning Rate = 0.001). Training was done in 10 epochs with batch treatment. After each epoch, the model was evaluated on the test set, and the best-performing version for future use (depending on the hit rate) was saved.

Evaluation

Two common matrices were used to evaluate the performance:

Hit Rate (HR): It was checked if the correct item has appeared in the top-k recommendations.

NDCG (Normalized Discounted Cumulative Gain): Measured the quality of the object's ranking and rewarded the correct elements that appeared more on the list.

A customized evaluation function calculated both matrixes for each model and produced comparative plots, to imagine how each one performed.

Replication of studies

To repeat the results:

- Download and decompress the dataset from provided links.
- Set up a Python 3.x environment and install the required libraries.
- Use the script or notebook feature to convert Json data to test.data format.
- Run the filtering and downsampling script to get the final dataset.
- Open Jupyter Notebook and run each section in order, from data preparation and model definitions to training and evaluation.

4. Experiments

Experimental setup and design

The purpose of our experiments was to evaluate the effectiveness of the matrix factor specifically to predict user preferences for children's books. We prepared it as a binary-classification problem (predicting whether a user interacts with a book) and a ranking problem (recommending top-N items).

We used a filtered and downsampled version of Goodreads Children Books Dataset. After preprocessing, the final dataset consisted of thousands of user-book-rating triples. To ensure reproducibility and fair evaluation, we used a leave-one-out split at the user level: for each user, we held out exactly one interaction as the test example.

To prevent data leakage, all user book interactions in the test set for users and books that appeared in the training set. This ensured our model was not making predictions for entirely unseen users or items.

Data collection and process

All the data used was already present and was downloaded from a public source. No additional scraping or user interaction was required. Treatment process includes:

Filtering raw JSON: Remove listings with missing data or rankings below 1.

Minimum limits: Excluding users with fewer than 5 ratings and books with fewer than 20.

Down sampling: Reduce data size by 20% of the post-filtered set to improve model iteration speed.

Train test Split: Strained to each user to appear in both sets, so the model allows user preferences to learn effectively.

These stages produced a clean, balanced data set with reliable user interaction patterns.

Conditions, control and variables

Independent variable

Model architecture: This was the primary variable that was examined with four different architectures: GMF, MLP, NeuMF and NeuMF transformer. Each represents another hypothesis on how user-aite interaction is the best modeling.

Embedding Dimensions (num_factors): For GMF, we used num_factors of 8, 16, 32 and 64 to inspect the effect of embedding size. For MLP, num_factors (for initial user/item embeddings) wasn32, which led to a input dimension of 64, which then passed by defined layers (eg [64, 32, 16, 8]). For NEUMF and NEUMF transformers, GMF component used num_factors = 16 and MLP component using num_factors = 64 based on GMF and MLP's best individual performance.

Pre-training Strategy (for NeuMF and NeuMF-Transformer): We explored whether pre-training the GMF and MLP components individually before combining them in NeuMF (and NeuMF-Transformer) offered performance benefits over training them from scratch within the combined architecture.

Dependent variable

Hit Rate (HR@K): The percentage of times the relevant item was found within the top-K recommendations.

Normalized Discounted Cumulative Gain (NDCG@K): A metric reflecting the quality of the ranking, giving more weight to relevant items found higher in the list.

Controlled variable

Dataset: All experiments were conducted on the same preprocessed, filtered, and downsampled test.data to eliminate data variability as a factor.

Evaluation Protocol: "Leave-one-out" strategy and negative testing samples (NUM_NG_NG_TEST = 99) corresponds to all models.

Optimization parameters: Adam Optimizer, Learning Rate (0.001) and BCE LOS features were all equal for training.

Batch size: A fixed batch_size of 256 was used for all training.

Maximum Epochs: All models were allowed to be trained in 10 epochs.

Random seeds: Consistent random seeds were used to ensure the reproducibility of random processes, such as negative sampling and initial weight randomization, in all experiments. This minimized variance is due to randomness.

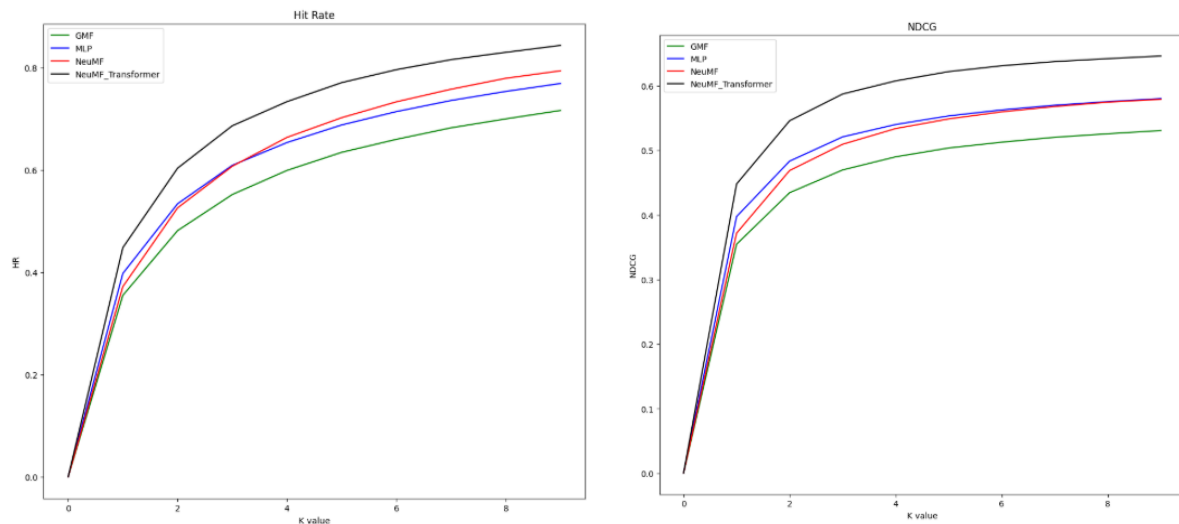
5. Results

This section presents empirical conclusions achieved from the experimental evaluation of various nerve collaboration models (NCF). The performance of each model architecture general Matrix Factorization (GMF), Multi-Lear Perceptron (MLP), Neural Matrix Factorization (NEUMF) and NUMF transformer (NUMF transformer) was assessed based on Hit Rate (HR) and Normalized Discounted Cumulative Gain (NDCG) across different values of K (the number of top recommendations).

Model Performance Overview

The training process for each model involves optimizing Binary Cross-Entropy loss over 10 epochs, with the best model checkpoint (based on test set Hit Rate) being saved. The final assessment was made on these stored optimal models.

Summary of Best Model Configurations and Their Performance (HR@10, NDCG@10)



General Matrix Factorization (GMF):

Experiments with the GMF model showed different performances based on the number of latent factors. The GMF16 configuration, using 16 latent factors, achieved the highest Hit Rate and NDCG between the continuous GMF variants. For example, GMF16 recorded an average HR@10 of approx. 0.729 and NDCG @10 at approx. 0.535. Performance usually increased K, which indicates a high probability of finding the object in question in a large recommendation list.

Multi-Layer Perceptron (MLP):

The MLP models, which leverage a multi-layer neural network to learn non-linear interactions, performed competitive performance. The MLP64 configuration, with its specific layer structure and initial embedding dimensions, achieved the best results for this architecture It achieved a mean

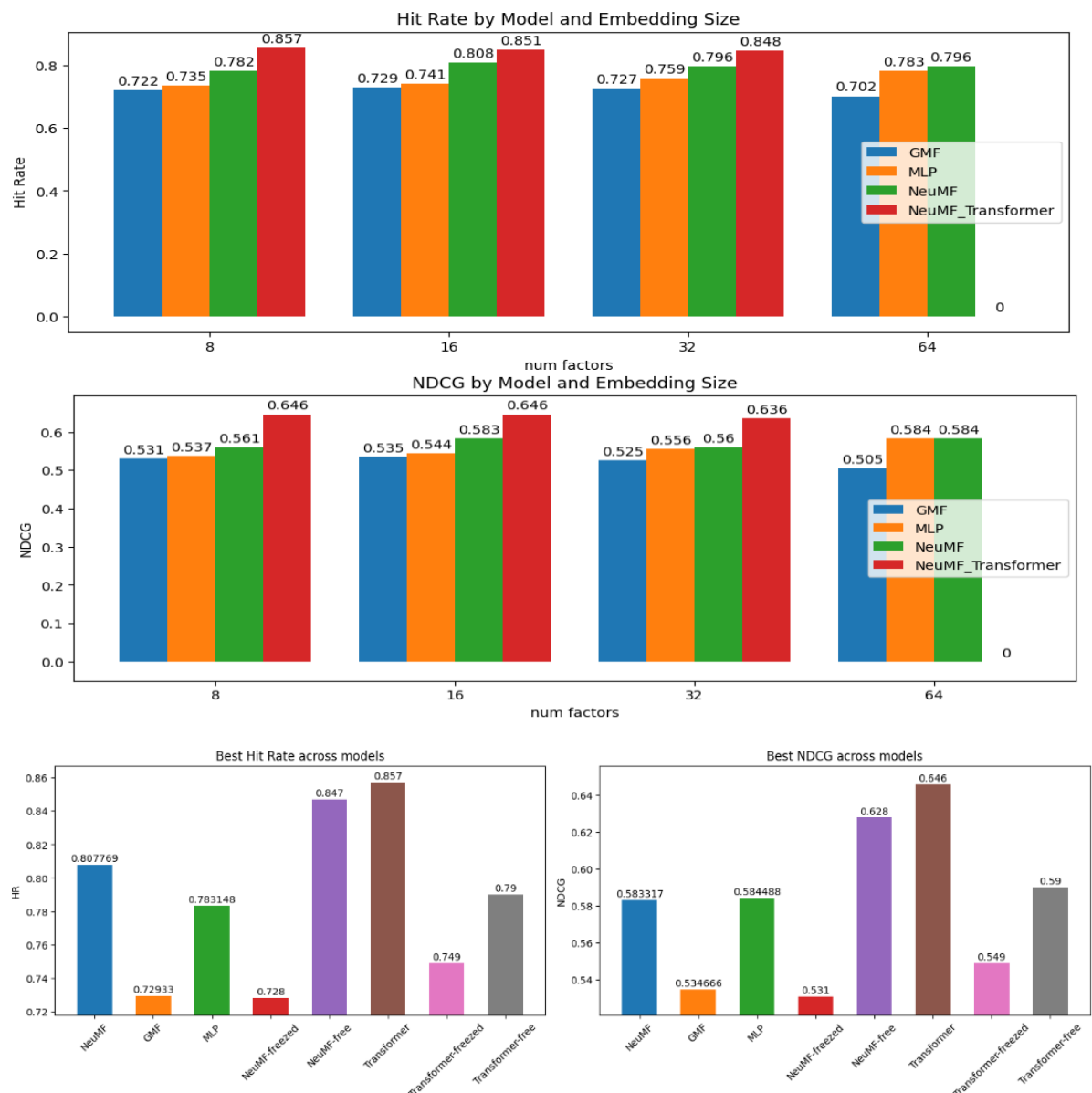
HR@10 of approximately 0.783 and an NDCG@10 of about 0.584. Similar to GMF, the performance metrics for MLP increased with higher K values.

Neural Matrix Factorization (NeuMF):

The NeuMF model, which combines the strength of GMF and MLP, continuously improved its individual components. NEUMF_CLASS16 Configuration (which mentions joint architecture where GMF used 16 factors and MLP used 64 factors) got an HR@10 of approx. 0.808 and NDCG@10 around 0.583.

Comparative Analysis Visualizations

The comparative view of all models (GMF16, MLP64, Neumf_Class16 and Neumf_Transformer32) was visually depicted through bar diagrams. These plots depicted an average Hit Rate and NDCG values for each model in a range of K values (1 to 10).



6. Discussion

The results of our experiments provide a strong insight into how different Neural Collaborative Filtering (NCF) models perform to handle the response to the recommendations from the book. Among the four tested models, NeuMF transforms provided the best general performance after NeuMF, MLP and GMF. As of GMF, which models the user-item ratio through simple linear interactions, the results showed that it can serve as a solid base, but its simplicity limits its ability to capture deep user preferences. MLP, which uses a nervous network on several levels to learn non-lecture interactions, performs better, suggests that adding non-linearity helps to highlight more complex patterns.

The biggest jump in the performance came with the NeuMF model. By combining the linear approach to GMF and non-linear power of MLP, NeuMF scored much better results in both ranking accuracy (NDCG) and recommendation relevance (Hit Rate). It confirms the advantage of combining different modeling strategies to achieve a more complete view of user behavior. NeuMF-Transformer took things a step forward. This model added a transformer encoder to the architecture, which possibly helped to capture deep or more abstract relationships between users and books, although our data did not include any clear sequence of interactions. The transformer's self-attention mechanism may have traveled on the hidden pattern that the traditional layers cannot easily detect. When we come to the transfer learning sections, if we keep the GMF and MLP layers frozen, in this case a worse performance was obtained than even the scratch NeuMF. However, if we get GMF and MLP layers free, in this case we saw a better improvement than even the best NeuMF, that is, as expected, the transfer learning application had negative results in the first case and increased the success of the model in the second case. However, it was not possible to say the same for the NeuMF_Transformer case. When our transfer learning application was frozen, it gave better results than the normal NeuMF frozen case. Even though there was a significant increase in the success rate of the model when it was free, it remained below expectations.

One important takeaway is the essential role of negative sampling when working with implicit feedback. Since we do not know which items users do not like, the models depend too much on the negative examples that are chosen well to learn. The success of all our models, especially more advanced, indicates that this strategy works well. For systems in the real world, which depend on implicit data, such as surfing history or streaming behavior, the use of a combination of linear and non-linear modeling as in NeuMF can produce a strong result and for those looking to further improve accuracy, and add attention mechanisms, such as in transformers, can offer an edge.

7. Conclusion

The project developed and evaluated various deep teaching models for children's book recommendation systems, specifically addressing the challenge with implicit feedback data. Our most important conclusions confirm that hybrid architectures do much better than their individual components. The Neural Matrix Factorization (NeuMF) model, combining GMF's linearity with non-linearity of MLP achieved a much higher Hit Rate and NDCG score compared to standalone GMF and MLP models.

In addition, the integration of an encoder into the NeuMF framework (NeuMF-Transformer) achieved the best general performance and performed a marginal but consistent improvement. This emphasizes the ability of an attention-based mechanism to capture more complex patterns in interactions between user-item, even in non-sequential contexts. Furthermore, when we applied transfer learning - loading pretrained GMF and MLP embeddings into NeuMF and NeuMF-Transformer - we observed that freezing both embedding layers led to worse performance than training from scratch, whereas allowing those pretrained embeddings to be fine-tuned consistently yielded the highest HR@10 and NDCG@10 for NeuMF. In the NeuMF-Transformer case, freezing the GMF and MLP layers still outperformed the frozen baseline NeuMF, but full fine-tuning did not deliver as large a boost as it did for plain NeuMF.

The importance of these results lies in providing a strong and effective approach to individual recommendations on the domains where clear negative feedback is rare. Our findings provide valuable insights into the fact that practitioners create recommendation systems for children's content, and help finding relevant and attractive books.

8. Contributors

Team member Umut Yıldırım played a role in the development of the first four steps of the project. He performed data preprocessing parts, created MLF and GMF models, created a baseline NeuMF model. He implemented transfer learning mechanism for NeuMF. Additionally, he created the transformer-based NeuMF model for attention system. He tested the created models according to different embeddings, implemented the plot visualization parts. Abrar Mustafa combined transformer-

based NeuMF with transfer learning applications, wrote the first 5 sections of the report. The remaining sections in the report were written by Umut Yıldırım.